

SREERAG E G

SDET & Automation Architect

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PROFESSIONAL SUMMARY

Performance-driven **SDET and Automation Architect** with a "Developer-First" approach to quality. Expert in engineering **distributed automation platforms** using **Python, FastAPI, and AWS (ECS/SQS)** to solve complex workflows in the **US Healthcare Insurance domain**. Specialized in building scalable, HIPAA-compliant frameworks that move beyond simple scripting into full-scale infrastructure orchestration.

CORE TECHNICAL STACK

- **Languages:** Python (Advanced OOP & Framework Design), SQL, JavaScript, Java.
- **Automation:** Playwright, Selenium, Web Scraping, PDF Data Extraction.
- **Backend & Orchestration:** FastAPI, Django, REST API Design, Async Processing.
- **Cloud & DevOps:** AWS (ECS, SQS, EC2, S3, Lambda), Docker, Git, CI/CD Pipelines.
- **Frameworks:** Page Object Model (POM), Error Classification & Self-Healing Logic, Retry Mechanisms.
- **Domain Specialization:** US Healthcare Insurance, Claims Processing, HIPAA Compliance, EOB/CMS-1500.

PROFESSIONAL EXPERIENCE

- Software Developer (SDET / Automation Architect)** | Kalki Safalya Consulting Services LLP *May 2025 – Present*
- **Architected a Distributed Automation Gateway:** Developed a high-performance orchestration layer using **FastAPI** to dynamically route automation tasks based on incoming JSON payloads.
 - **Scalable Infrastructure Deployment:** Deployed a containerized worker pool on **AWS ECS**, utilizing **AWS SQS** as a message broker to handle asynchronous claim submissions at scale.
 - **Performance Engineering:** Designed a multi-portal automation platform that increased insurance claim submission speed by **40%** while maintaining production-grade reliability.
 - **Reliability & Compliance:** Implemented automated error classification, screenshot evidence capture for audits, and sensitive data masking to maintain strict HIPAA compliance.
 - **Internal Tooling:** Built claims tracking systems using **Pandas** and integrated browser automations with enterprise Rails backends via message queues.
- Full Stack Developer (Intern)** | Aspire Systems *Nov 2024 – May 2025*
- **API Engineering:** Developed and optimized REST APIs using **Django Rest Framework (DRF)** and MySQL, ensuring seamless data flow for enterprise applications.
 - **Quality Assurance:** Led automated API documentation using **Swagger** and participated in Agile sprints, focusing on code reviews and modular system design.
 - **Auth Systems:** Implemented secure authentication flows, including Google OAuth and CRUD operations for internal portals.
- Freelance Python Developer (Automation)** *May 2024 – Nov 2024*
- **Legacy Transformation:** Converted one-off Selenium scripts into maintainable, framework-compliant code for insurance web applications.
 - **Third-Party Integration:** Integrated external APIs to optimize backend performance and automation reliability.

KEY PROJECTS

- **Insurance Claim Web Automation:** Automated US insurance form submissions featuring CAPTCHA handling, robust logging, and reusable base classes for scalable portal interaction.
- **AI-Powered Image Recognition:** Developed a classification system with **95% accuracy** using TensorFlow and OpenCV.
- **AVI (App for Visually Impaired):** Android app featuring real-time object detection and audio feedback.

EDUCATION & CERTIFICATIONS

- **B.Tech in Computer Science and Engineering** | Christ College of Engineering (7.85 CGPA).
- **Certifications:** Data Engineering with PySpark, IBM Python for Data Science, AI For Everyone (Andrew Ng), Django for Web Development.